Stroke

CLINICAL AND POPULATION SCIENCES

Identification of Clinically Relevant Brain Endothelial Cell Biomarkers in Plasma

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BACKGROUND: Proteins expressed by brain endothelial cells (BECs), the primary cell type of the blood-brain barrier, may serve as sensitive plasma biomarkers for neurological and neurovascular conditions, including cerebral small vessel disease.

METHODS: Using data from the BLSA (Baltimore Longitudinal Study of Aging; n=886; 2009–2020), BEC-enriched proteins were identified among 7268 plasma proteins (measured with SomaScanv4.1) using an automated annotation algorithm that filtered endothelial cell transcripts followed by cross-referencing with BEC-specific transcripts reported in single-cell RNA-sequencing studies. To identify BEC-enriched proteins in plasma most relevant to the maintenance of neurological and neurovascular health, we selected proteins significantly associated with 3T magnetic resonance imaging–defined white matter lesion volumes. We then examined how these candidate BEC biomarkers related to white matter lesion volumes, cerebral microhemorrhages, and lacunar infarcts in the ARIC study (Atherosclerosis Risk in Communities; US multisite; 1990–2017). Finally, we determined whether these candidate BEC biomarkers, when measured during midlife, were related to dementia risk over a 25-year follow-up period.

RESULTS: Of the 28 proteins identified as BEC-enriched, 4 were significantly associated with white matter lesion volumes (CDH5 [cadherin 5], CD93 [cluster of differentiation 93], ICAM2 [intracellular adhesion molecule 2], GP1BB [glycoprotein 1b platelet subunit beta]), while another approached significance (RSPO3 [R-Spondin 3]). A composite score based on 3 of these BEC proteins accounted for 11% of variation in white matter lesion volumes in BLSA participants. We replicated the associations between the BEC composite score, CDH5, and RSPO3 with white matter lesion volumes in ARIC, and further demonstrated that the BEC composite score and RSPO3 were associated with the presence of ≥1 cerebral microhemorrhages. We also showed that the BEC composite score, CDH5, and RSPO3 were associated with 25-year dementia risk.

CONCLUSIONS: In addition to identifying BEC proteins in plasma that relate to cerebral small vessel disease and dementia risk, we developed a composite score of plasma BEC proteins that may be used to estimate blood-brain barrier integrity and risk for adverse neurovascular outcomes.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: blood-brain barrier ■ cerebral small vessel disease ■ dementia ■ endothelial cells ■ neuroimaging ■ proteomics ■ single-cell gene expression analysis

Brain endothelial cells (BECs) are the primary cell type of the blood-brain barrier (BBB), where their tight junctions lining the neurovasculature enable them to function as unique mediators of brain homeostasis, including metabolism, inflammation, and neuronal functioning.[1] BBB dysfunction (eg, abnormal permeability, impaired glucose transport, etc) is a common feature of neurodegenerative, cerebrovascular, and neuroinflammatory diseases,

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Nonstandard Abbreviations and Acronyms

AD	Alzheimer disease
ARIC	Atherosclerosis Risk in Communities
BBB	blood-brain barrier
BEC	brain endothelial cells
BLSA	Baltimore Longitudinal Study of Aging
CMHs	cerebral microhemorrhages
CSF	cerebral spinal fluid
CSVD	cerebral small vessel disease
GFAP	glial fibrillary acidic protein
LIs	lacunar infarcts
NfL	neurofilament light
PDGFR-ß	platelet-derived growth factor receptor beta
pTau-181	phosphorylated Tau 181
WMLs	white matter lesions

such as multiple sclerosis.[2] BBB dysfunction has been causally implicated in many of these diseases and is now considered a viable therapeutic target.[3,4] Although intrathecal levels of a BEC protein (CDH5 [cadherin 5]) and a pericyte protein (PDGFR-ß [platelet-derived growth factor receptor beta]) correlate with cognitive performance and Alzheimer disease (AD) biomarkers,[5,6] there remains a critical need for cost-effective, scalable, noninvasive and clinically relevant biomarkers of BBB integrity that can identify individuals at risk for neurological conditions, such as cerebral small vessel disease (CSVD).

CSVD is a form of cerebrovascular pathology associated with increased risk of stroke, vascular dementia, and AD. CSVD is commonly defined using structural neuroimaging parameters, including white matter lesion volumes (WMLs), cerebral microhemorrhages (CMHs), and lacunar infarcts (LIs).[7] Although the pathogenesis of this disease remains actively debated, BBB endothelial dysfunction is hypothesized to be a primary mechanism.[8] Several studies have linked certain inflammatory proteins measured in blood with CSVD structural neuroimaging abnormalities, yet there is insufficient evidence to suggest such biomarkers are preferentially expressed by BECs, pericytes, or other components of the BBB.[9] Thus, there is an unmet need to identify plasma biomarkers of BBB integrity that can predict the downstream clinical effects of BBB dysfunction or increased BBB permeability.

Using data from the BLSA (Baltimore Longitudinal Study of Aging), proteins specifically expressed by BECs were identified among 7268 plasma proteins using an automated annotation algorithm that filtered endothelial cell transcripts from the Human Protein Atlas followed by cross-referencing with BEC-specific transcripts reported in single-cell RNA-sequencing (RNA-seq) studies. To identify BEC-specific proteins in plasma most relevant to neurovascular health, we selected proteins significantly associated with 3T magnetic resonance imaging (MRI)–defined WMLs. We then examined the predictive validity of these candidate BEC biomarkers by relating them to WMLs, CMHs, and LIs in the ARIC study (Atherosclerosis Risk in Communities). Finally, we examined if these candidate BEC biomarkers, when measured during midlife, were related to dementia risk over a 25-year follow-up period.

METHODS

Data Availability Statement

The data that support the findings of this study are available from the corresponding author on reasonable request.

Study Sample

This study used data from the BLSA, which measures longitudinal physical and cognitive outcomes among community-dwelling volunteers[10] (Figure 1). The BLSA protocol was approved by the institutional review board of the National Institute of Environmental Health Sciences, National Institutes of Health (03AG0325), and informed consent was obtained for all participants. 3T-MRI scans were collected in 2009-2010. Blood samples were collected at the time of MRI scan. Participants with history of neurological conditions that could affect brain structure or function (eg, strokes, seizures, brain surgery) and participants with missing data were excluded (ie, complete case analysis). Reporting follows STROBE (Strengthening the Reporting of Observational Studies in Epidemiology) reporting guidelines.

3T-MAGNETIC RESONANCE IMAGING

Using a 3T Philips Achieva scanner, magnetization-prepared rapid gradient echo (repetition time=6.8 ms, echo time=3.2 ms, flip angle=8°, image matrix=256×256×170, voxel size=1 mm×1 mm×1.2 mm) and fluid-attenuated inversion recovery (repetition time=11 s, echo time=68 ms, inversion time=2800 ms, image matrix=240×240×150, voxel size=0.83 mm×0.83 mm×3 mm) scans were collected. To segment WMLs, fluid-attenuated inversion recovery scans along with MUSE (Multi-Atlas Region Segmentation Utilizing Ensembles)[11] segmented magnetization-prepared rapid gradient echo scans were used with a convolutional deep neural network (DeepMRSeg).[12]

Plasma Proteomics

Proteins were measured using the SomaScan v4.1 platform, which contained 7288 SOMAmer reagents targeting 6399 annotated human protein targets. Blood samples were collected using standardized protocols and plasma was frozen at −80 °C until analysis. Using 102 blind duplicates, proteins with intraassay coefficients of variation >50% were excluded (n=20). The median intraassay coefficients of variation was 4.5% for all measured proteins; coefficients of variation for

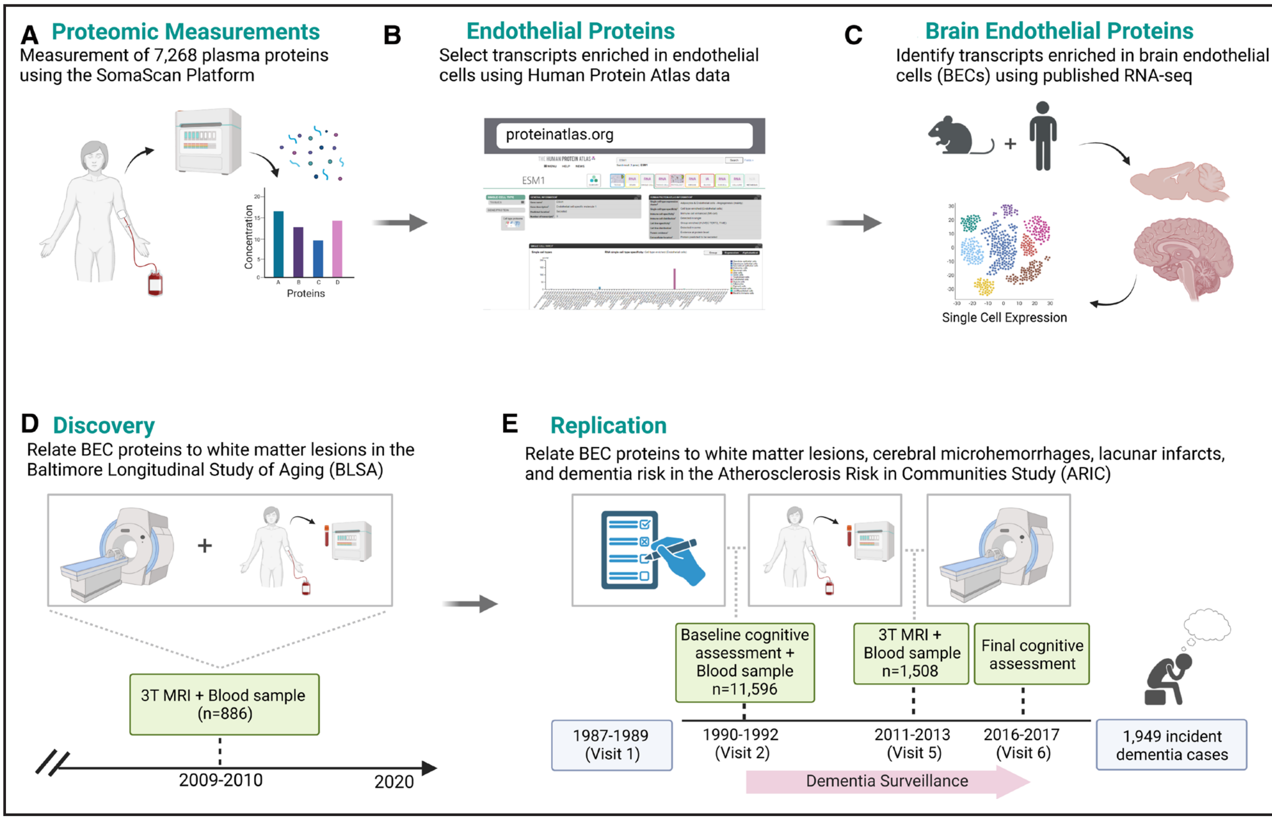

Figure 1. Overview of study design.

BEC indicates brain endothelial cell; MRI; magnetic resonance imaging; and RNA-seq, RNA-sequencing. Created with BioRender.com.

BEC candidate biomarkers were <6%.[13] Protein values were $\log_2$ transformed and those beyond 5 SD from the mean were winsorized. AD and related dementias (ADRD) biomarkers were measured using the Single Molecule Array (Simoa) Neurology 4-Plex-E and pTau-181 (phosphorylated Tau 181; V2) assays on the HD-X instrument (Quanterix). Assays were run in duplicate, and values were averaged. Intraassay coefficients of variations were 2.8, 1.9, 5.0, 5.1, and 4.4% for $A\beta$ (amyloid beta)$_{40}$, $A\beta_{42}$, GFAP (glial fibrillary acidic protein), NfL (neurofilament light), and pTau-181, respectively. Standardized values and $A\beta_{42/40}$ ratio were used in analyses.

Endothelial Cell–Enriched Transcripts

To identify endothelial-enriched transcripts, an automated annotation algorithm was applied to single-cell transcriptomic data sourced from the Human Protein Atlas. For the corresponding genes of each SomaScan protein target, the algorithm aligned single-cell transcript levels (normalized transcripts per million) from 76 unique cell types in the Human Protein Atlas database to quantify relative expression of each target (*Pc*) in comparison to its mean expression and distribution across all cell types (Equation 1). A specificity factor (*Z*) of 2 was used to identify transcripts expressed in endothelial cells at levels >2 SD above mean expression across cell types. To account for the exponential nature of protein expression between varying cell types, geometric means and SDs were used.

$$P_c > \mu_{gP}\sigma_{gP}{}^{Z} \quad (1)$$

Equation 1: Relative expression of protein *P* in cell type *c*: P_c : consensus transcript expression levels (normalized transcripts per million) of protein *P* in cell type c; μ_{gP} : geometric mean of protein *P* across all cell types; σ_{gP} : geometric SD of protein *P* across cell types; *Z*: specificity factor.

BEC-Enriched Transcripts

To identify proteins preferentially expressed by BECs, we cross-referenced transcripts identified as endothelial specific (see above) with empirically defined BEC-specific transcripts reported in 5 human and 6 mouse single-cell and single-nuclei RNA-seq analyses.[14–23]

Cerebral Spinal Fluid BBB Biomarkers

We used corresponding plasma and cerebral spinal fluid (CSF) SomaScan v4.1 proteomic measurements derived from 18 control and 18 AD participants made available by the Emory Goizueta Alzheimer's Disease Research

Center[24] to relate plasma BEC proteins to established measures of BBB integrity: CSF PDGFRB, CSF fibrinogen, and CSF/plasma-albumin ratio.[2]

ARIC Study

The ARIC study is a community-based cohort study that initially enrolled 15 792 adults (ages 45–65 years) between 1987 and 1989. Informed consent was obtained for all participants and protocols were approved by institutional review boards at participating centers. Proteins were measured using the SomaScan v4.0 platform with plasma from blood samples collected at visits 2 and 5 using standardized protocols[25] (Figure 1). SomaLogic protein quality control steps and 3T-MRI acquisition procedures in ARIC have been described previously.[25,26] Quantitative assessment of fluid-attenuated inversion recovery images using a computer-aided segmentation program was used to determine WMLs; CMHs and LIs were identified and confirmed by trained imaging technicians and radiologists, as described previously.[26] Dementia was classified between visits 2 and 6. Dementia diagnosis was adjudicated through cognitive assessment tests, telephone screening, informant ratings, hospital records, and death record review (Supplemental Methods).[25]

BEC Composite Scores

Using BEC proteins that were related to WMLs in BLSA and available on both SomaScan's v4.1 and v4.0 platforms, a weighted composite score was calculated. BEC protein weights were beta estimates (β) obtained from unadjusted linear regression models that included WMLs as the outcome (*F*) and WML-related proteins (*P*) as the predictors (Equation 2a). The composite score (*Cw*) was the Log_2 transformed sum of Log_2 protein abundance (*P*) calibrated according to such weights (β; Equation 2b). Protein-specific beta estimates were derived from the BLSA and applied to the same proteins measured in ARIC to test the predictive performance of the composite score across studies.

$$F\ (x)\ =\ \beta_1 P_1 + \beta_2 P_2 + \beta_3 P_3 \tag{2a}$$

$$C_w = log_2\,(\beta_1 P_1 + \beta_2 P_2 + \beta_3 P_3) \tag{2b}$$

Equation 2—Weighted composite score of BEC proteins. Equation 2a: *F(x)*: association of white matter lesion volumes with Log_2 BEC protein abundance (*P*) in an unadjusted linear regression model. Equation 2b: *Cw*: Log_2 transformed sum of weighted proteins *P*; *Pβ*: product of Log_2 protein abundance (*P*) weighted by *P*s respective β; β: coefficient derived from Equation 2a. relating protein $P_{1\text{-}3}$ to WML volume in BLSA.

Covariates

Age (years), sex (male/female), race (White/non-White), and education level (years) were defined based on participant reports at the time of study visit. *APOEε*4 status (0 *ε*4 alleles/≥1 *ε*4 alleles/missing) was defined via polymerase chain reaction with restriction isotyping using the TypeIIP enzyme Hhai or Taqman method. Estimated glomerular filtration rate (eGFR)–creatinine was defined at the time of blood sample collection using the CKD-EPI (Chronic Kidney Disease Epidemiology Collaboration) criteria.[27] Comorbid diseases were defined using a comorbidity index calculated as the sum (score range, 0–8; converted to percentage to account for missing data) of 8 conditions: obesity, hypertension, diabetes, cancer, ischemic heart disease, chronic heart failure, chronic kidney disease, and chronic obstructive pulmonary disease.[28]

Statistical Analyses

Linear regression models were used to examine the associations of BEC-specific proteins with WMLs and plasma ADRD biomarkers. Models were adjusted for age, sex, race, education, *APOEε*4, eGFR-creatinine, total white matter volume (WML analyses), and the comorbidity index. Spearman correlations were used to relate BEC-specific proteins to CSF BBB biomarkers. In the ARIC sample, candidate BEC biomarkers were related to WML using linear regression and CMHs and LIs using logistic regression. These models were adjusted for age, sex, race-center, education, *APOEε*4, eGFR-creatinine, total intracranial volume (WML analyses), and cardiovascular risk factors (BMI, diabetes, hypertension, and current smoking status). To determine how candidate BEC biomarkers relate to 25-year dementia risk, Cox models adjusting for the aforementioned covariates were used. The Cox proportionality assumption was tested by computing and plotting Schoenfeld residuals. WMLs, GFAP, NfL, and pTau-181 values were log2 transformed to correct for skewness. K.A. Walker has full access to all the data in the study and takes responsibility for its integrity and analysis. Statistical significance was defined at 2-sided $P<0.05$. Analyses were performed in R 4.2.

RESULTS

Identification of BEC Proteins

Of the 7268 SomaScan protein targets, 6390 were annotated by the automated algorithm that sourced single-cell transcript data from the Human Protein Atlas, and 48 (0.75%) of these were identified as endothelial cell enriched (Equation 1; Figure 2A; Table S1). Of these, 28 were previously reported as BEC-specific using single cell and single nuclei RNA-seq (Figure 2B; Table S2). Cell-specific transcripts reported for other cell types (eg, pericytes, astrocytes, perivascular fibroblasts) were absent from our list of BEC transcripts (Table S2).

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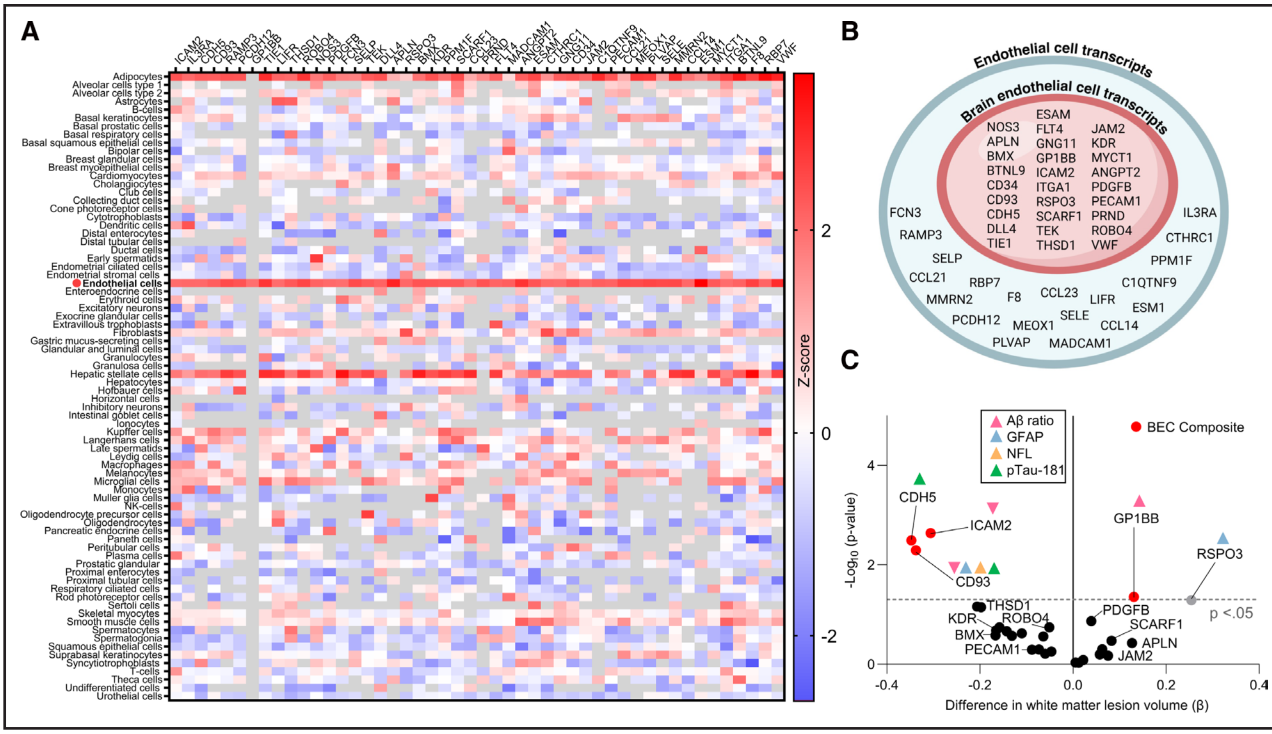

Figure 2. Discovery of brain endothelial cell (BEC) proteins in plasma associated with cerebral small vessel disease. **A**, Heatmap showing the identification of endothelial cell–enriched transcripts using an automated annotation algorithm of Human Protein Atlas expression data across 76 cell types. **B**, Venn diagram displaying a subset of transcripts empirically defined as BEC-specific in ≥1 single cell and single nuclei RNA-sequencing (RNA-seq) studies. **C**, Volcano plot showing the associations of BEC proteins as well as a composite score of BEC proteins with white matter lesion volumes and Alzheimer disease and related dementias (ADRD) biomarkers. Results derived from linear regression models adjusted for age, sex, race, education, APOEε4, estimated glomerular filtration rate–creatinine, and an index of 8 comorbid conditions (obesity, hypertension, diabetes, cancer, ischemic heart disease, chronic heart failure, chronic kidney disease, chronic obstructive pulmonary disease); white matter lesion analyses additionally adjusted for total white matter volumes. βs reflect adjusted differences in white matter lesion volumes per each log2 increase in protein level. Red dots indicate statistical significance (unadjusted $P<0.05$), gray dots indicate marginal significance (unadjusted $P=0.051$), and black dots indicate no statistical significance (unadjsuted $P>0.05$). Significant positive and negative associations with ADRD biomarkers are indicated by up and down arrows, respectively. Aβ ratio indicates amyloid $\beta_{42/40}$; CD93, cluster of differentiation 93; CDH5, cadherin 5; GFAP, glial fibrillary acidic protein; GP1BB, glycoprotein 1b platelet subunit beta; ICAM2, intracellular adhesion molecule 2; NfL, neurofilament light; pTau-181, phosphorylated Tau 181; and RSPO3, R-Spondin 3.

Association of Plasma BEC Proteins With CSVD and ADRD Biomarkers

Next, we examined whether levels of the 28 BEC-specific proteins measured in plasma were associated with WML volume in 886 BLSA participants (age, 66 years [SD=15]; 55% female; 66% White; Table S3; Figure S1). Among the 28 plasma proteins we identified as BEC-specific, higher levels of CDH5, CD93 (cluster of differentiation 93), and ICAM2 (intracellular adhesion molecule 2) were related to decreased WMLs (protective proteins), whereas elevated GP1BB (glycoprotein 1b platelet subunit beta) was associated with increased WMLs (pathogenic protein; Figure 2C; Figure 3A through 3D). Another protein, RSPO3 (R-Spondin 3), showed a positive association with WMLs that approached significance ($P=0.051$; Figure 2C; Figure 3E). Associations with WML volume remained significant after false discovery rate correction for ICAM2, CDH5, and CD93 (Table S4). Given their evidence of BEC-specificity and their associations with CSVD, we nominated these 5 proteins as candidate BEC biomarkers.

To better understand whether candidate BEC biomarkers may be used to quantify BBB disruption, we related candidate plasma proteins to established measures of BBB integrity (CSF PDGFRB, CSF fibrinogen, and CSF/plasma-albumin ratio).[2] While CDH5 was associated with lower CSF fibrinogen level ($\rho=-0.53$; $P<0.001$), indicative of preserved BBB integrity, GP1BB was positively associated with CSF:plasma-albumin ratio ($\rho=0.43$; $P=0.01$), indicative of a more permeable BBB (Table S5). Plasma ICAM2, CD93, and RSPO3 were unrelated to CSF measures of BBB integrity.

Next, we created a BEC composite score comprised of 3 candidate BEC biomarkers available on both SomaScan's v4.1 and v4.0 platforms (CDH5, CD93, RSPO3). A weighted score was calculated as the sum of BEC protein levels multiplied by their respective β estimates obtained from a protein-only linear regression model with WML volume as the outcome variable (Equation 2a and 2b; Table S6A). The BEC composite score was

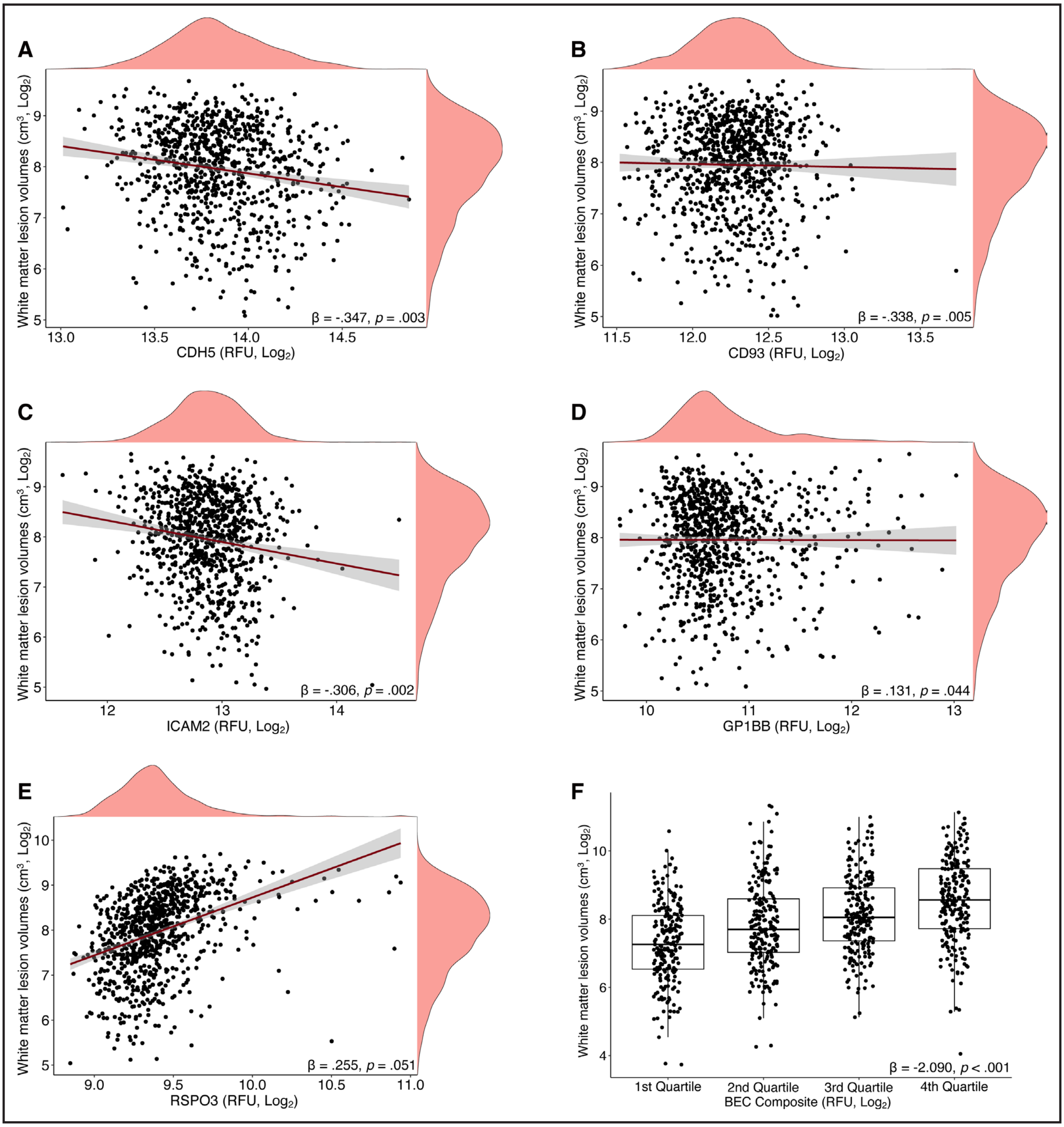

Figure 3. Associations of candidate brain endothelial cell (BEC) biomarkers and a BEC protein composite score with white matter lesion volumes in BLSA (Baltimore Longitudinal Study of Aging) participants.
Scatterplots and lines of best-fit showing predicted changes in white matter lesion volumes associated with (**A**) CDH5 (cadherin 5), (**B**) CD93 (cluster of differentiation 93), (**C**) ICAM2 (intracellular adhesion molecule 2), (**D**) GP1BB (glycoprotein 1b platelet subunit beta), and (**E**) RSPO3 (R-Spondin 3). Results (βs, *P* values) display predicted white matter lesion volumes per each log2 increase in protein level derived from linear regression models adjusted for age, sex, race, education, APOEε4, estimated glomerular filtration rate–creatinine, total white matter volumes and an index of 8 comorbid conditions. **F**, Stratified boxplot illustrating changes in white matter lesion volumes according to BEC protein composite score quartiles. Results (β, *P* values) were derived from unadjusted linear regression models. RFU indicates relative fluorescence units.

significantly associated with WMLs (β=2.09; *P*<0.001), accounting for 10.8% of the variation in WML volume in unadjusted models (Figure 3F). After adjusting for demographic variables, *APOEε*4, kidney function (eGFR-creatinine), and a comorbidity index, the BEC composite score remained significantly associated with WML volume (β=0.73; *P*<0.001); the full model accounted for 48.6% of the variation in WML volume (Table S6B). A covariate-only model accounted for 47.5% of variation.

To further assess the clinical and pathological relevance of BEC biomarkers, we examined their levels in relation to plasma ADRD biomarkers (Table S3; Figure S2). Higher

CD93 was associated with lower $A\beta_{42/40}$, and increased pTau-181, GFAP, and NfL, whereas higher ICAM2 and GP1BB levels were associated only with lower and higher $A\beta_{42/40}$, respectively. Higher CDH5 was related to elevated pTau-181, and higher RSPO3 was associated with elevated GFAP (Figure 2C). The BEC composite score was not related to ADRD biomarkers (Tables S3 and S7).

External Validation of Plasma BEC Proteins

To determine the predictive validity of BEC-specific proteins with respect to MRI markers of CSVD, we used data from 1508 ARIC participants (age, 76 years [SD=5]; 59% female; 75% White) with available 3T-MRI data (Table S3; Figure S3). Three of our 5 BEC candidate biomarkers were measured in ARIC (CDH5, RSPO3, and CD93). Consistent with the results from our discovery (BLSA) analysis, lower CDH5, and higher RSPO3 were significantly associated with greater WML volume (Figure 4A). Higher RSPO3 was also associated with greater odds of having ≥1 CMHs (Figure 4A; Table S8A and S8B). CD93 was not associated with ARIC neuroimaging measures, and no proteins were associated with the presence of LIs (Figure 4A; Table S8C). The BEC composite score,

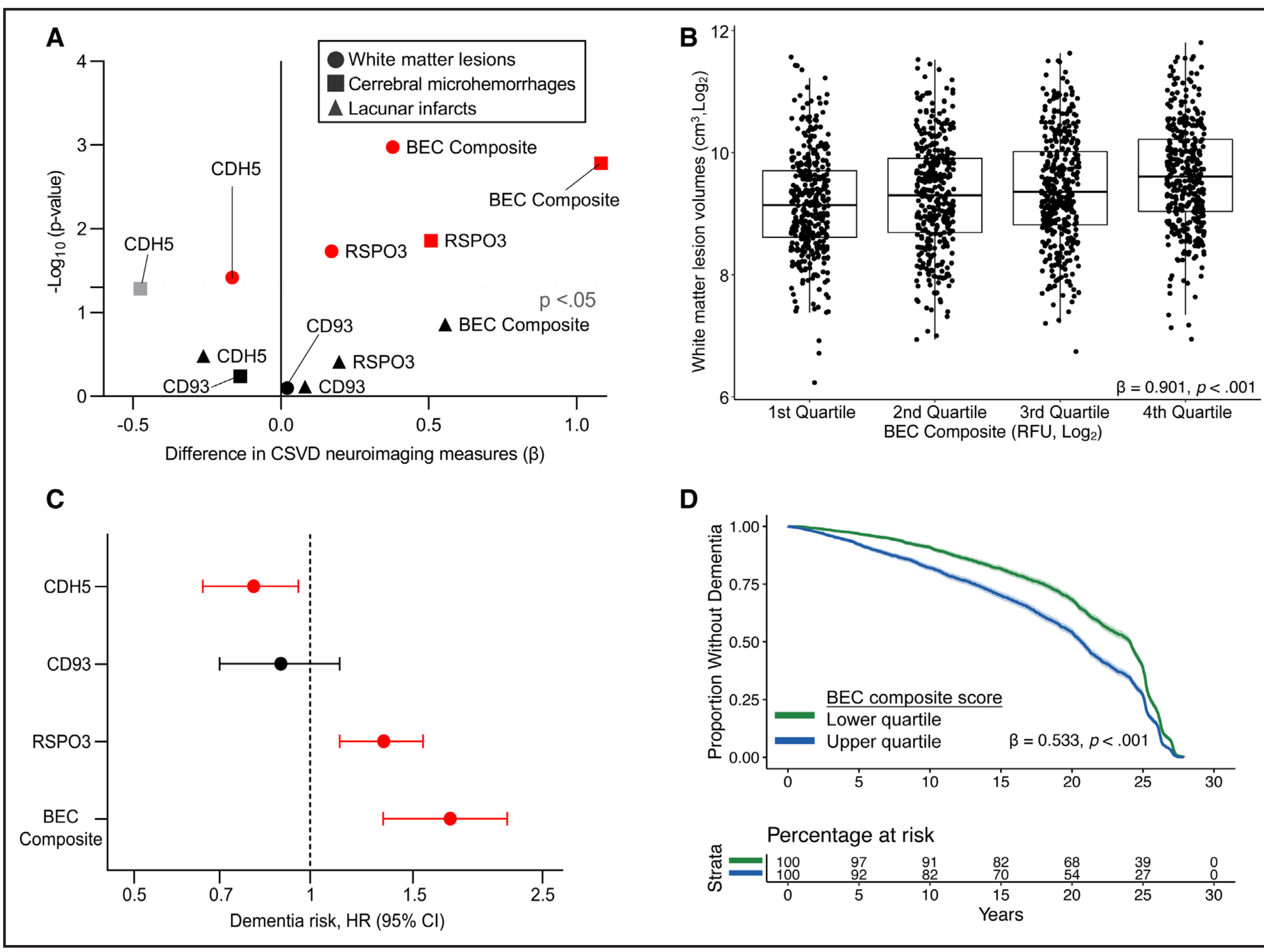

Figure 4. External validation of candidate brain endothelial cell (BEC) biomarkers and a BEC composite score using clinically and pathologically relevant outcome measures.

A, Volcano plot showing the associations of BEC proteins and the BEC composite score with matter lesion volumes, cerebral microhemorrhages, and lacunar infarcts. Results derived from linear or logistic regression models adjusted for age, sex, race-center, education, APOEε4, estimated glomerular filtration rate (eGFR)–creatinine, total intracranial volume, and cardiovascular risk factors (body mass index, diabetes, hypertension, and current smoking status). βs reflect adjusted differences in white matter lesion volumes or cerebral microhemorrhage/lacunar infarct odds per each log2 increase in protein level. Red markers indicate statistical significance (unadjusted $P<0.05$), gray markers indicate marginal significance (unadjusted $P=0.052$), and black markers indicate no statistical significance (unadjsuted $P>0.05$). **B**, Stratified boxplot illustrating changes in white matter lesion volumes according to BEC composite score quartiles. Results (βs, *P* values) were derived from unadjusted linear regression models. **C**, Forest plot showing 25-year dementia risk associated with candidate BEC biomarkers and the BEC composite score. Hazard ratios reflect adjusted differences in dementia risk per each log2 increase in BEC protein and BEC composite score levels. Red indicates statistical significance (unadjusted $P<0.05$). **D**, Kaplan-Meier curves (and a corresponding at-risk table) displaying cumulative risk of incident dementia over 25 y according to bottom and upper quartiles that reflect log2 increases in BEC composite score. Dementia risk results (hazard ratios [HRs], βs, *P* values) were derived from Cox proportional hazards regression models adjusted for age, sex, race-center, education, APOEe4, eGFR-creatinine, and cardiovascular risk factors. CD93 indicates cluster of differentiation 93; CDH5, cadherin 5; CSVD, cerebral small vessel disease; RFU, relative florescence units; and RSPO3, R-Spondin 3.

derived using weights (β estimates) from the BLSA discovery cohort, was significantly associated with WML volume (β=0.90; P<0.001) in ARIC, accounting for 3.6% of the WML volume variation in unadjusted models (Figure 4B). After adjusting for demographic variables, *APOEε4*, kidney function (eGFR-creatinine), and cardiovascular risk factors, the BEC composite score remained significantly associated with WML volume (β=0.38; P=0.001); the full model accounted for 23.7% of the variation in WML volume (Table S8D). A covariate-only model accounted for 22.4% of variation. The BEC composite score was also associated with the presence of CMHs and LIs, although the relationship with LIs was no longer significant after covariate adjustment (Figure 4A; Table S8E and S8F).

Having demonstrated an association between BEC candidate biomarkers, a BEC composite score, and MRI-defined CSVD in ARIC, we next used a sample of 11 596 nondemented ARIC participants to examine whether these same biomarkers, measured during midlife, were associated with dementia risk over a 25-year follow-up period (baseline age, 57 years [SD=6]; 56% female; 77% White; Table S3; Figure S4). Higher CDH5 and RSPO3 were associated with decreased and increased dementia risk, respectively, consistent with their directions of associations with neuroimaging measures (Figure 4C; Table S9A). CD93, on the other hand, showed a nonsignificant inverse associated with dementia. In an unadjusted model, each SD increase in the BEC composite score was associated with a 168% increase in 25-year dementia risk (hazard ratio, 2.68 [95% CI, 2.13–3.38]; Table S9B). The association between the BEC composite score and dementia risk was attenuated, yet remained highly significant, in a model adjusted for aforementioned covariates (hazard ratio, 1.70 [95% CI, 1.33–2.17]; Figure 4C; Table S9B). The 25-year dementia event rate was 15.4% for participants in the first BEC composite score quartile (10-year survival rate:91.0%). By contrast, participants in the top BEC composite score quartile (suggesting poorer BEC function/integrity) had a 25-year dementia event rate of 18.8% (10-year survival rate:82.0%; Figure 4D). A summary of evidence for BEC candidate proteins is provided in Figure 5.

DISCUSSION

Optimal BBB functioning is essential to neurological well-being, yet there remains a critical need for cost-effective, scalable, and noninvasive BBB biomarkers to monitor neurovascular health and identify individuals

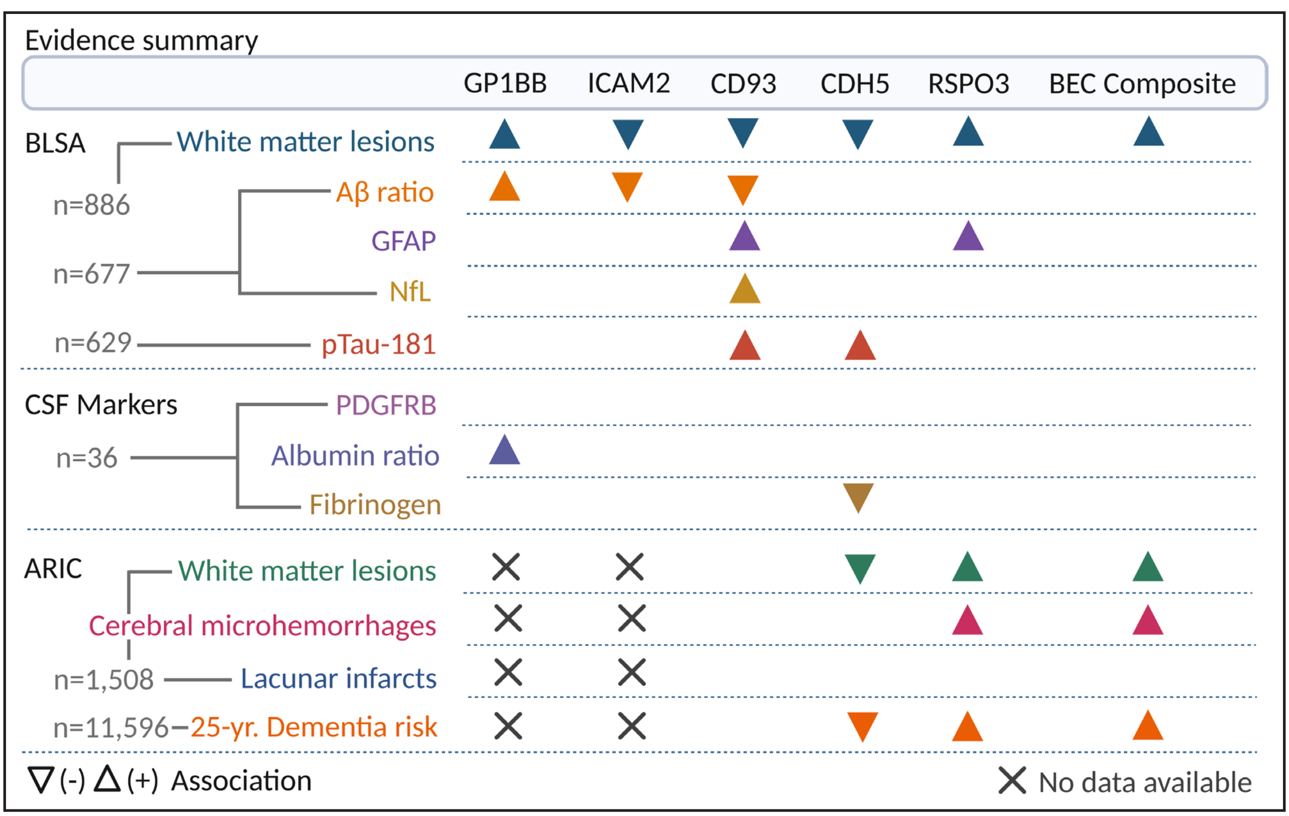

Figure 5. Summary of findings from the current study.

ARIC indicates Atherosclerosis Risk in Communities; BEC, brain endothelial cell; BLSA, Baltimore Longitudinal Study of Aging; CD93, cluster of differentiation 93; CDH5, cadherin 5; CSF, cerebral spinal fluid; GFAP, glial fibrillary acidic protein; GP1BB, glycoprotein 1b platelet subunit beta; ICAM2, intracellular adhesion molecule 2; NfL, neurofilament light; PDGFRB, platelet-derived growth factor receptor B; and RSPO3, R-Spondin 3. Created with BioRender.com.

at risk for conditions such as CSVD. After selecting 28 proteins preferentially expressed by BECs, the primary cell type of the BBB, we identified a subgroup of BEC proteins (CDH5, CD93, ICAM2, GP1BB, RSPO3) that were associated with WML volume, a marker of CSVD, in a community sample of older adults. We demonstrated that plasma CDH5 and GP1BB levels were associated with biomarkers of BBB integrity (CSF fibrinogen and CSF/plasma-albumin ratio, respectively), and we used BEC-specific proteins to create a composite score that accounted for 11% of variation in WML volume. With data from an external cohort, we replicated the WML volume associations for 2 of the 3 available BEC proteins (CDH5 and RSPO3) and the BEC composite score, and demonstrated that these 2 proteins were also associated with 25-year dementia risk (findings summarized in Figure 5).

Several candidate BEC biomarkers associated with protective effects in our analyses (CDH5, CD93, ICAM2) are transmembrane glycoproteins implicated in endothelial tight junction permeability as well as the translocation of leukocytes and macrophages.[29–31] As these cell adhesion molecules can be upregulated in the context of cellular stress (eg, ischemia, hypoxia),[32–34] the inverse associations between these proteins and white matter burden could suggest the maintenance of BBB integrity or attenuated BEC interactions with infiltrating peripheral immune cells. Conversely, given its function in arterial platelet recruitment following vascular injury,[35] the increased WMLs associated with GP1BB could suggest underlying damage to neurovasculature. As RSPO3 regulates vascular network remodeling and activates inflammatory gene expression that can exert synergistic effects on BBB permeability,[36–38] the adverse neuroimaging and clinical outcomes associated with RSPO3 may reflect secondary consequences of aberrant inflammatory signaling (a postulation supported by its positive relationship with plasma GFAP), compromised BBB integrity or systemic drivers of angiogenesis, such as differential metabolic demands.[39] With the exception of RSPO3, our candidate BEC proteins displayed unexpected associations with ADRD biomarkers, including $A\beta_{42/40}$ and pTau-181. Given that BBB integrity likely influences the extent to which biomarkers originating from the central nervous system can migrate into the periphery, the observed relationships may be attributed to neurobiological mechanisms that are distinct from coevolving neurodegenerative processes.[40]

Given its established role in the assembly and maintenance of endothelial tight junctions, its replicated relationship with WML volumes, and association with CSF fibrinogen and dementia risk, plasma CDH5 (also known as vascular endothelial cadherin) appears to be a top candidate biomarker of endothelial/BBB integrity. Beyond the nomination of individual protein biomarkers, our BEC composite score adds to recent studies that have distilled proteomic panels into multi-protein biomarkers capable of predicting neurological conditions associated with BBB dysfunction.[41,42] Although the BEC composite score accounted for only 11% and 4% of WML volume in BLSA and ARIC in unadjusted models, respectively, the finding that it was independently associated with white matter burden across cohorts and 25-year dementia risk indicates its utility as a measure of neurological health. Although it accounted for limited variation (≈1%) in WML volume beyond that provided by other covariates in fully adjusted models (which included age and a comorbidity index of hypertension, obesity, diabetes, etc,), this unique predictive variance significantly accounted for differences in WML burden and dementia risk. Together, these findings suggest that the BEC composite score (1) largely captures variation in WML volume and CSVD-related outcomes linked to known risk factors such as age and other cardiovascular conditions and (2) predicts a small portion of clinically relevant information that is not otherwise captured by age or the other known risk factors measured in the present study. As BEC integrity is implicated in numerous neurological conditions, such as stroke, epilepsy, and brain tumors,[1,2] our candidate biomarkers and the BEC composite score should also be examined as potential markers of cerebrovascular health in each of these disease states. To enhance the applicability of our current findings, validation in larger, more heterogeneous cohorts and additional mechanistic investigations (eg, in vivo knock-down/overexpression studies) are needed, as are longitudinal studies that can track BEC biomarkers across the spectrum of disease progression.

The current study exhibited several strengths, including an innovative approach to identify BEC-specific proteins using high-throughput expression data, the use of 3T-MRI across 2 large, multiracial community samples, and a multidecade follow-up of dementia incidence. However, several limitations should be considered. First, according to Human Protein Atlas data, many endothelial-specific proteins were also highly expressed in hepatic stellate cells and adipocytes; while such profiles could represent legitimate coexpression patterns, these may also reflect imperfect extraction methods that failed to completely limit contamination from endothelial cells themselves before RNA-seq, given that such cell types are routinely cocultured in vitro and highly innervated in vivo.[43] Although our 2-step selection process enhanced the likelihood of identifying proteins preferentially expressed by endothelial cells of brain origin, non-brain endothelial cells may have also contributed to their plasma levels, especially if such cells are involved in the pathogenesis of central nervous system and non-central nervous system diseases. Relatedly, although our methods accurately excluded cell-specific markers of other cell types, our identification of BEC-specific proteins relied on transcriptomic data that may have failed to fully capture differences in protein levels. Finally, the protein measurements used in our study are reliable,[44] but it's possible that biological fluctuations (eg, diurnal) may have hindered our capacity to detect associations with some proteins. Despite these limitations, our analyses identified BEC plasma biomarkers and a BEC protein composite score that relate to CSVD and dementia risk, suggesting their potential utility in distinguishing individuals at risk for adverse neurovascular outcomes.

ARTICLE INFORMATION

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Disclosures

None.

Supplemental Material

Supplemental Methods
Tables S1–S9
Figures S1–S4

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